Numerical and Algebraic Expressions — Distributing and Combining Without Breaking the Rules

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Two rules do most of the work on this page, and breaking either one is the most common way to lose an expression problem.

- **Distributing** means multiplying *every* term inside the parentheses — including the sign in front of each one.
- **Like terms** have exactly the same letter part. $5x$ and $2x$ are like; $5x$ and $3y$ are not, and neither are $5x$ and 5.
- **The substitution test:** if two expressions are really equivalent, they give the same value for every number you put in. One disagreement proves they are different expressions.

1. Write $4(x + 7)$ without parentheses. Multiply every term inside by 4.

Answer: ________________

2. **Find and fix.** Kim writes $3(x + 5) = 3x + 5$.
Test both expressions at $x = 4$. Kim's version gives 17. Work out the value of $3(x + 5)$ at $x = 4$ and write it on the answer line, then say in a few words which term Kim forgot to multiply.

Answer: ________________

3. Find and fix. Devon writes $5x + 3y = 8xy$.
Test both expressions at $x = 2$ and $y = 3$. Devon's version gives 48. Work out the value of $5x + 3y$ at those values and write it on the answer line, then say in a few words why $5x$ and $3y$ cannot be pulled into one term.

Answer: ________________

4. Simplify by combining like terms only:
$7a + 4 - 2a + 9$

Answer: ________________

5. Find and fix. Simplify $8 - 2(x - 3)$.
Nadia writes $8 - 2x - 6$, which is $2 - 2x$, and at $x = 1$ her version gives 0.
The -2 must be multiplied by *both* terms in the parentheses. Work out the correct value of $8 - 2(x - 3)$ at $x = 1$ and write it on the answer line.

Answer: ________________

6. Simplify completely:

$5(2x - 3) + 4x$

Distribute first, then combine only the terms that are alike.

Answer: ________________

7. Factor $6x + 15$ by taking out the greatest common factor. Then check your answer by distributing it back — the check should return $6x + 15$ exactly.

Answer: ________________

8. Explain, then simplify.

(a) A classmate insists that $5x + 3y$ "must be able to become one term." Explain, in your own words, why two unlike terms cannot be combined. Remember that a letter stands for a number nobody has told you yet.

(b) Simplify $3(2x-4)-2(x-5)$, then check your simplified expression by substituting $x = 3$ into it. Write that value on the answer line. (A classmate who keeps both signs inside the second bracket gets -10 here.)

Answer: __________________